

HASBY FAHRUDIN

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SKILLS

Agentic | Machine Learning | Computer Vision | Cloud Architecture | Backend Development | Project Management
Python | C++ | JavaScript | HTML | CSS | Docker | Git | FastAPI | Django | .NET | JIRA | Github Actions | Teraform
TensorFlow | PyTorch | Scikit-learn | Transformers | LangChain | spaCy | Milvus | Pandas | OpenCV | AWS

EXPERIENCE

Senior AI Engineer Crosscert, Inc

Mar 2025 - Present
Seoul, South Korea

- Implemented Langgraph-based multi-agent system for progressive query disambiguation and multi-step RAG techniques, doubling technical support agent resolution rates.
- Optimized the RAG knowledge entry pipeline with a dynamic chunking strategy and SentenceTransformer-based semantic chunking, boosting adaptability of agent knowledge formats.
- Refined agent knowledge integrity via metadata-driven refactoring of unstructured data using Unstructured.
- Improved long-conversation coherence by developing a context-aware conversational memory system leveraging vector embedding-based query relevancy with agentic retrieval.
- Managed a Django+Next.js-based platform to orchestrate customer order workflows & automated SSL provisioning.

AI Engineer AIBrain, Inc

May 2023 - Mar 2025
Seoul, South Korea

- Developed a DAG-based multi-agent orchestration framework enabling advanced persona modeling, adaptive fallback strategies, and context-aware dynamic action routing, optimized for self-hosted small foundational model.
- Built an interruptible async streaming service with asyncio to support concurrent multi-user LLM responses, reliably stopping generation on client disconnects and improving system stability.
- Transformed conversational agent sessions from a 3-user concurrent stateful setup to a scalable, Dockerized, stateless framework on AWS ECS, enabling virtually unlimited concurrency.
- Drove data-driven features in SAIWA, a VR football match analysis platform, delivering predictive insights and performance analytics that enabled pilot programs with Bundesliga teams.
- Architected a football pass analysis pipeline, leveraging XGBoost for predictive metrics and applying game-theoretic modeling to evaluate decision-making and passing strategies.
- Scaled video analysis pipeline with AWS MQ-based queue processing, expanding capacity to virtually unlimited user.
- Achieved 7th place in SoccerNet 2023 using a heuristic tracklet aggregation pipeline with ESRGAN super-resolution, pose-based localization, and confidence-weighted digit recognition.
- Led a cross-functional lean engineering team (3-7 members) across VR, Web, and AI stacks.

AI Engineer Crosscert, Inc

May 2021 - May 2023
Seoul, South Korea

- Built a multi-agent imitation learning model for optimal defensive positioning in football, leveraging spatial-temporal player tracking data.
- Developed a video post-processing pipeline combining polynomial regression and heuristic corrections, reducing spatial-temporal errors and lowering bug incidence by 60%.
- Recovered and documented legacy VR, ML, and platform codebases, restoring project continuity and reducing onboarding time for new engineers.
- Implemented a fast artifact detection system for football broadcast videos using feature-based pattern recognition, matching baseline accuracy with lower compute and data needs.
- Developed a PoC 3D GAN-based system to transform NFT assets into diverse artistic styles and extrapolated 3D representations, enhancing visual richness and interaction potential.

Master's Research Assistant Wireless Network Lab, SeoulTech

Mar 2020 - Sep 2022
Seoul, South Korea

- Achieved ~10% accuracy improvement in MAML for few-shot head movement prediction in VR tasks, optimizing computation time with Meta-Curvature and Layer-Specific Learning Rates (LSLR).
- Improved real-time VR head movement prediction accuracy by ~12% using an ensemble learning approach.
- Benchmarked meta-learning techniques across diverse datasets, creating a reference guide on their application.

Software Engineer IDEMIA

Jun 2019 - Feb 2020
Jakarta, Indonesia

- Enhanced system reliability for Remote OS Library Loader leading to 30% decrease in production issues.
- Ensure 100% delivery time for software testing task

- Developed a lightweight facial emotion classifier using hybrid feature extraction, achieving ~70% accuracy with real-time prediction on edge devices.
- Engineered an assistive device for visually impaired users to perceive facial expressions, achieving 79% accuracy in real-world testing.

EDUCATIONS

Seoul National University of Science and Technology
M. Sc., Electrical and Information Engineering

Mar 2020 - Sep 2022
Seoul, South Korea

- Thesis: Combining Improvements in Model-Agnostic Meta-Learning.
- GPA: 4.25 of 4.50.

Institut Teknologi Sepuluh Nopember
B. Eng., Electrical Engineering

Aug 2015 - Jul 2019
Surabaya, Indonesia

- Thesis: Deep learning based facial expressions recognition system for assisting visually impaired persons.
- GPA: 3.3 of 4.0

HIGHLIGHTED PROJECTS

SAIVA: Sport AI Virtual Assistant

[\[link\]](#)

- Designed to give teams and athletes a competitive edge, SAIVA is the brainchild of a collaboration between AI and VR development experts, professional football coaches, and sports scientists.

Orkes: A lightweight orchestration framework for LLM agents

[\[link\]](#)

- A lightweight orchestration framework for LLM agents, no abstractions you don't need, no bloated APIs, just clean coordination logic that works.

Facex: Lightweight, High-Performance Facial Expression Classifier

[\[link\]](#)

- Python library for detecting faces and classifying emotions in images lightweight, efficient threading and object pooling for concurrent processing making it suitable for high-performance application.

PUBLICATIONS

SoccerNet 2023 Challenges Results
CVPR 2023

[\[link\]](#)

- Contributing on proposing pipe-line to extract jersey number of each player on football match broadcast video.

Deep Learning Based Facial Expressions Recognition System for Assisting Visually Impaired persons
Bulletin of Electrical Engineering and Informatics 2020

[\[link\]](#)

- Developed a deep learning-based facial expression recognition system on a wearable device to help visually impaired individuals interpret facial expressions during communication

OTHERS

- Peer-reviewed papers for **CVSport** 2024 and **ICANN** 2024.
- Represented SAIVA at **Sports Innovation** 2024 (Düsseldorf, Germany) and **VIVATECH** 2024 (Paris, France).
- **Speaker** Regional Scrum GatheringSM Seoul 2024, Enhancing Product Management with ChatGPT